KARTIKEYA S BAGALORE

711 Superior Ct, Naperville IL 60563 • kartikbagalore@gmail.com • +1-331-226-3783

SUMMARY

Staff Applied Research Scientist in the Core – LLM group at ServiceNow, working on Agentic workflows and Large Language Models to build, evaluate and deploy agentic evaluation frameworks and coding sidekick applications.

TECHNICAL SKILLS

Programming: Python, Java, R, Spark, SQL **ML & LLM:** PyTorch, HuggingFace (PEFT, TRL), vLLM, WandB, Docker **Collaboration:** Git, Confluence, JIRA, Sphinx

PROFESSIONAL EXPERIENCE

Staff Applied Research Scientist • ServiceNow Inc., Chicago*Jan 2024 – Present*

- Agent-Bench: Leading Agentic Evaluation research at ServiceNow. Designed and built 2 enterprise benchmarks to evaluate frontier and open-source LLMs across ITSM, HR, and CSM domains — measuring task completion, instruction following, and quality of responses.
- Key contributor to NowAssist, a developer tool to generate ServiceNow code in the native code editor. Finetuned the LLM on ServiceNow data using PEFT and RLHF to align results according to best practices. Used HuggingFace (peft, trl) and PyTorch to perform LoRA finetuning on Open-source models for internal usage; experiments, data and models logged in WandB.
- Built an LLM tuning framework for reproducibility and comparability of experiments — now the de facto platform for training and evaluating all LLM experiments.
- Took over deployment responsibilities, building and deploying pre- & post-processing pipelines around the model. Deployment stack consists of a vLLM backend hosted on a Triton server.

Senior Applied Research Scientist • ServiceNow Inc., Chicago*Jan 2021 – Jan 2024*

- Agent Zero: Using word embeddings (transformer-based models & sentence encoders) to identify and automatically resolve high volume-low complexity tickets, leading to expedited closure of 20–25% of tickets across customers.
- Built a ServiceNow app for annotation of text using a low data approach and a clustering-based technique using HDBSCAN. Speeds up labeling by 5x–7x by identifying difficult samples for labels.

Data Scientist • ServiceNow Inc., Chicago*Sept 2018 – Jan 2021*

- Knowledge Gap resolution: built robust ML solutions to identify issues which may have preexisting solutions in the Knowledge base in real-time.
- Contribution to the Content Summarization and Content Extraction applications, along with assistance in performance testing, building integrated tests, and debugging ML related issues while supporting clients.

Data Scientist • Sears, Chicago*Jan 2018 – Sept 2018*

- Owner of the Price Elasticity of Demand (PeD) estimation model, estimating product price elasticities and providing optimal prices for weekly promotions using a predictive model trained on 3 years of historical sales data via Stan on GCP.

Co-op Data Scientist • Micron Technology, Inc., Washington D.C.*Mar 2017 – Dec 2017*

- Contributed to creating and maintaining Python packages in PyPI and conda used for authentication, data retrieval, analysis and reporting of Chip & Wafer images, z-plot features & usage metrics.

EDUCATION

Clemson University, Clemson SC — Master of Science in Electrical Engineering, Dec 2016
Graduate Coursework: Artificial Neural Networks, Machine Learning & Pattern Recognition, Statistical & Probability Theory.
Master's Thesis: Detection & Classification of Epileptic Spikes in Patient EEG using HMMs and LPC Features.